



Netramind

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1 PURPOSE

The purpose of this Standard Operating Procedure (SOP) is to establish the requirements for electronic records, electronic signatures, and compliance with the US Food and Drug Administration (FDA) requirements in 21 CFR Part 11, and current global data integrity requirements.

This SOP provides a standard to govern the use and control of electronic records and electronic signatures within computerized systems.

2 SCOPE

This SOP applies to all Good Clinical Practice (GCP) computerized systems that are used by Netramind that generate, maintain, and transmit electronic records and/or electronic signatures in support of clinical trials.

3 REFERENCES

- SOP-014 Computer System Validation
- SOP-004 GxP Records Retention
- SOP-003 GxP Training Program and Curricula
- FDA Guidance for Industry: Part 11, Electronic Records; Electronic Signatures – Scope and Application; 21 CFR Part 11 Electronic Records and Electronic Signatures
- ICH E6(R3): Guidelines for Good Clinical Practice (GCP), Section 5.5.3

4 DEFINITIONS

- 4.1 Computerized Systems Validation (CSV)—A process of establishing and documenting that the specified requirements of a computerized system can be consistently fulfilled from design until decommissioning of the system or transition to a new system. The approach to validation should be based on a risk assessment that takes into consideration the intended use of the system and the potential of the system to affect trial participant protection and the reliability of trial results
- 4.2 Data Acquisition Tool (DAT)—A paper or electronic tool designed to collect data and associated metadata from a data originator in a clinical trial according to the protocol and to report the data to the sponsor. The data originator may be a human (e.g., the participant or trial staff), a machine (e.g., wearables and sensors), or a computer system from which the electronic transfer of data from one system to another has been undertaken (e.g., extraction of data from an electronic health record or laboratory system). Examples of DATs include but are not limited to CRFs, interactive response technologies, clinical outcome assessments, including patient-reported outcomes (PROs) and wearable devices, irrespective of the media used.
- 4.3 Data Integrity—Data integrity includes the degree to which data fulfill key criteria of being attributable, legible, contemporaneous, original, accurate, complete, secure and reliable such that data are fit for purpose.



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- 4.4 Digital Signature—An electronic signature based on cryptographic methods of originator authentication, computed by using a set of rules and parameters such that the identity of the signer and the integrity of the data can be verified.
- 4.5 Electronic Record —Any combination of text, graphic, data, audio, pictorial, or other information representations in digital form that is created, modified, maintained, archived, retrieved, or distributed, by a computer system. A system transaction or data set is not considered to be an electronic record until the data entry is committed to a database.
- 4.6 Electronic Signature —A computer data compilation of any symbol or series of symbols executed, adopted, or authorized by an individual to be legally binding, legally equivalent to the individual's handwritten signature.
- 4.7 Good Clinical Practice (GCP)— A standard for the planning, initiating, performing, recording, oversight, evaluation, analysis, and reporting of clinical trials that provides assurance that the data and reported results are reliable and that the rights, safety and well-being of trial participants are protected.
- 4.8 Hybrid System—An environment consisting of both electronic and paper-based records.
- 4.9 Metadata—The contextual information required to understand a given data element. Metadata is structured information that describes, explains, or otherwise makes it easier to retrieve, use, or manage data to allow for the appropriate evaluation of the trial conduct.
- 4.10 Signature—A unique mark, symbol, or entry executed, adopted, or authorized by an individual, in accordance with applicable regulatory requirements and/or practice to show expression of will and allow authentication of the signatory (i.e., establish a high degree of certainty that a record was signed by the claimed signatory). A signature may be physical or electronic
- 4.11 Source Records— Original documents or data (which includes relevant metadata) or certified copies of the original documents or data, irrespective of the media used. This may include trial participants' medical/health records/notes/charts; data provided/entered by trial participants (e.g., electronic patient-reported outcomes (ePROs)); healthcare professionals' records from pharmacies, laboratories, and other facilities involved in the clinical trial; and data from automated instruments, such as wearables and sensors.
- 4.12 Trial Participant Identifier— A unique identifier assigned to each trial participant to protect the participant's identity and used in lieu of the participant's name when the investigator reports adverse events and/or other trial-related data.

5 RESPONSIBILITIES

5.1 All employees



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5.1.1 Accept that their e-signature is the legally binding equivalent of their handwritten signature.

5.2 System Owner (SO) and Information Technology (IT)

5.2.1 Performs a complete Computer System, GxP, and 21 CFR Part 11 assessment of the entire computer systems environment to identify compliance gaps, developing a plan to bring applications into compliance, and assisting with remediation efforts.

5.3 Quality Assurance (QA)

5.3.1 Conducts audits of electronic records and audit trails to monitor functionality and/or determine if a breach has occurred.

5.4 Regulatory Affairs (RA)

5.4.1 If Netramind submits regulatory documents directly to regulatory agencies using electronic signatures, Regulatory Affairs ensures submission of required certifications (e.g., Letter of Non-Repudiation) as applicable.

6 PROCEDURE

6.1 Certify Use of E-signatures to the Regulatory Agency

6.1.1 This section applies only when Netramind submits documents directly to regulatory authorities using electronic submission systems (e.g., ESG).

6.1.2 RA submits a Letter of Non-repudiation on behalf of Netramind to FDA, stating that e-signatures used in the Electronic Submissions Gateway (ESG) are intended to be the legally binding equivalent of handwritten signatures (**TMP-015-01, Letter of Non-Repudiation Template**).

6.1.2.1 Users requesting a new WebTrader ESG user account upload an electronic copy of Letter during account registration via the FDA ESG User Portal (<https://esgportal.fda.gov/>)

6.1.2.2 Users with active ESG accounts email the Letter to the ESG Help Desk (ESGHelpDesk@fda.hhs.gov).

6.1.3 Training records demonstrating user understanding of electronic signature accountability are maintained in accordance with the GxP Training SOP.



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6.2 Risk Management for Computerized Systems

- 6.2.1 System Owner (SO) and Information Technology (IT) personnel use the **FRM-015-01 21 CFR Part 11 Assessment Form** to review Netramind systems to determine if each system is GxP controlled. If a system is found to be not GxP relevant, the form is archived in IT and QA files to document the assessment.
- 6.2.2 For GxP relevant computerized systems, Netramind SO and IT personnel determine if they meet 21 CFR Part 11 requirements using the **FRM-015-01 21 CFR Part 11 Assessment Form**. If the system-specific is unable to meet the requirements, gaps identified during the initial validation or periodic assessment are addressed as follows:
- 6.2.2.1 The SO and IT identify the system-specific gaps and the potential impact based on the intended use. The method of mitigation for the gaps is documented and procedural controls are set.
- 6.2.2.2 The SO and IT periodically review risks and controls to ensure continued compliance.
- 6.2.2.3 The SO and IT recommend removing the system from use for all GxP activities and data collection.
- 6.2.3 Validation of Systems
- 6.2.3.1 The SO and IT validate computer systems according to the approved procedure or plan to ensure the system meets intended use. Validation is considered part of the system development lifecycle (SDLC).
- 6.2.3.2 The SO and IT conducts changes to computerized systems following an approved change control procedure. A detailed description of the system is produced and kept current. The extent of the validation is determined on intended use of the system and in accordance with **SOP-014 Computer System Validation**.
- 6.2.4 Ability to Generate Accurate and Complete Copies of Records in Both Human Readable and Electronic Form.
- 6.2.4.1 Accurately print electronic data or display the electronic data on screen along with copies of electronic data obtained from the system.
- 6.2.5 To protect records, the SO and IT:
- Adhere to formalized back-up recovery procedures.



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- Secure data by physical and/or electronic means to maintain data integrity.
- Develop and implement controls to prevent, detect, and mitigate harmful effects of computer viruses, worms, malware, and other harmful codes on systems and records.
- Follow data back-up and archival procedures at defined intervals.
- Perform periodic checks of disaster recovery and business continuity planning.
- Provide limited access to Authorized Individuals only.
- Use passwords on systems to ensure access control is limited to authorized individuals. Access to a system is only provided after obtaining appropriate approvals.
- Periodically review user access (e.g., annually or as defined by procedure) to verify if there were any breaches.
- Change passwords on a periodic basis.
- Ensure passwords are of appropriate length and complexity to ensure that systems are protected at all times.
- For software hosted by a third party (i.e., Software as a Service), ensures that contractual agreements for back up and disaster recovery are in effect and monitored.

6.2.6 For the use of Secure, Computer-Generated, Time-Stamped Audit Trails, IT and the SO:

- Secure the audit trails stored on systems and ensure audit trails can be read by authorized individuals. Ensure audit trails contain the date, time, identity of operator, and operation type.
- Ensure visibility in the audit trail of modified or deleted, old or new data values. Write-access to the system tables or files containing audit trail information is limited to authorized personnel.
- Archive and back up audit trails with the associated data.
- Establish procedural controls to ensure changes to GxP data are recorded and tracked if the system does not contain an audit trail.

6.2.7 Use of Operation System Checks

- 6.2.7.1 On a periodic basis, the SO and/or IT perform operational system checks where sequencing of steps is potentially critical to GxP data. In such cases, the sequence may be identified in the appropriate requirement or design documents. It is acceptable to verify correct sequencing via validation testing or by procedure.



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6.2.7.2 When a continuation check is required, it may be done by a second individual or by validated electronic means.

6.2.8 Use of Authority Checks

6.2.8.1 The SO and IT ensure that only authorized individuals may use the system, electronically sign a record, access the computerized system, alter a record, and perform operations using authority checks.

6.2.8.2 The SO and IT enable system security to ensure that system access is controlled through passwords.

6.2.8.3 The SO and IT train users who have the appropriate user rights/ access roles to conduct the specified business activities on the system.

6.2.9 System Training

6.2.9.1 The SO and IT follow a documented training process ([SOP-003 GxP Training Program and Curricula](#)) to train users on their respective GxP computerized systems prior to use.

6.2.10 Accountability

6.2.10.1 All employees accept that their e-signature is the legally binding equivalent of their handwritten signature. Each person involved in electronically signing ERs must have documented training that demonstrates understanding of this statement and must understand that they are responsible for their actions initiated under e-signatures.

6.2.11 Control of System Documentation

6.2.11.1 The SO and IT ensure that revision, distribution, and access to operation, maintenance, and validation system documentation are controlled by approved procedures.

6.3 Electronic Signatures (ES) Management

The SO and IT ensure that Netramind systems align with current regulatory requirements for signatures:

6.3.1 Signature Manifestations, Signature, and Record Linking



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6.3.1.1 Signed e-signatures contain information associated with the signing that clearly indicates the following:

- Full name of the signatory
- Date and time the signature was executed
- Meaning (ex: review, approval, responsibility, or authorship) associated with the signature

6.3.1.2 Signatures are required to be kept and linked with their associated e-records.

6.3.1.3 E-signatures associated with the signed ERs can be displayed or printed, and include the following:

- The full name of the signatory along with the date and time
- Date and time the signature was executed, and the meaning or role associated with the signature

6.3.1.4 The time stamp must reflect the time zone in which the e-signature was applied.

6.3.2 E-Signature General Requirements

6.3.2.1 Unique to One Individual—Username/ password combination may only be associated with one individual. An individual's username/ password is deactivated upon leaving Netramind.

6.3.2.2 Netramind Verifies Employee Identities—Establishes a validation to verify the identity of the individual assigned an account associated with an e-signature.

6.3.3 E-Signature Components and Controls

6.3.3.1 Employees must use at least two (2) distinct identification components (applicable to non-biometric signatures). These components are used only by the genuine owner and administered in such a way that anyone other than the genuine owner requires collaboration between two or more individuals.

- Netramind employs unique username/ password combinations as representations of e-signatures, and only the owner assigned to the unique username/ password may use the username/ password.

6.3.3.2 The signature components apply to a single period of controlled system access as well as when a user logs out between controlled system access activities.



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- When an individual executes a series of signings during a single, continuous period of controlled system access, the first signing is executed using all e-signature components (username and password). Subsequent signings are executed using at least one (1) electronic signature component.
- When an individual executes one or more signings not performed during a single, continuous period of controlled system access, each signing is executed using all the e-signature components.

6.3.3.3 E-Signatures based on biometrics are secure.

- Biometrics are permitted to be used at Netramind. If biometrics are used, validation of the biometrics per intended use is required.

6.3.4 Controls for Identification (ID) Codes and Passwords

6.3.4.1 Maintaining Uniqueness of the ID Code or Password

- Deactivate all users upon leaving Netramind.
- Ensure the system does not permit creation of users with the same credentials.

6.3.4.2 Periodic Review of ID Codes and Passwords

- Password change requirements are defined by procedure or by a system capable of enforcing changes.

6.3.5 Controls for Open Systems

6.3.5.1 The controls outlined above also apply to open systems. In addition, open systems used to create, modify, maintain, or transmit ERs employ procedures and controls designed to ensure authenticity, integrity, and confidentiality of ERs from the point of their creation to the point of their receipt.

6.3.5.2 When critical data is transmitted over an open system, data protection measures are required (ex: digital signature, data encryption) to maintain data integrity.

6.4 Electronic Records (ER) Management

6.4.1 Electronic records defined in this SOP are subject to the applicable requirements of **SOP-004 GxP Records Retention**.



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6.4.2 The following GxP records (structured and unstructured) are considered equivalent with applicable record protection measures:

6.4.2.1 Examples of structured records includes, but not limited to the following:

- Database records
- Metadata or audit trails

6.4.2.2 Examples of unstructured records include, but not limited to, the following:

- Text documents
- Images

6.4.3 GxP personnel scan paper records into a modifiable image format (ex: PDF) and maintain within the appropriate Functional Department record repositories. These records may be considered source documentation to support GxP-related activities. Paper records may be used when electronic records are not available or practical.

6.4.3.1 All regulatory authority retention requirements apply to the content of the records, but not the format. If a retention requirement applies to a paper record, it also applies to an electronic copy of that record.

6.4.3.2 Reviews the record to ensure legibility and completeness of the converted record. The converted ER does not require an electronic signature when the documents contain the original signatures.

6.4.3.3 Retains converted records in a controlled environment. The controlled environment includes controlled personnel access and print capabilities.

6.4.3.4 Only destroys the existing paper record after following the procedures in **SOP-004 GxP Records Retention**.

7 ATTACHMENTS

- **FRM-015-01 21 CFR Part 11 Assessment Form**
- **TMP-015-01 21 Letter of Non-Repudiation Template**



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8 REVISION HISTORY

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SIGNATURE PAGE

[ADD ADDITIONAL ROWS AS NEEDED]

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